Disambiguating “Layer” Across the Trilogy: Four Layer Concepts (Instinct, DNA, Action, Reasoning) Consistently Defined Across Papers—Paper 3 Adds Cross-Perimeter Governance (Reasoning Layer Can Cross; Instinct Layer Cannot)

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Date: May 15, 2026

Attribution

This work derives from and formalizes commitments across the CKS theory trilogy: “The Instinct/Reasoning Separation Outside the Model” (Li, April 2026), “Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems” (Li, April 2026), and “Inter-Self Coordination via Shared Substrate / Full Aspect Integration” (Li, April 2026). It does not introduce new axioms. Its sole contribution is to establish, in operational form, the precise and consistent meanings of four “layer” terms across the three papers—instinct layer, DNA layer, action layer, and reasoning layer—and to record the one extension Paper 3 makes to those definitions: the cross-perimeter governance property that specifies which layers may cross the inter-Self perimeter during a coordination event.

Abstract

The CKS trilogy uses four “layer” terms—instinct layer, DNA layer, action layer, and reasoning layer—across all three papers. Readers encountering the terms across papers face two sources of confusion. First, the terms appear without explicit cross-paper inventory in any single location, which can create the false impression that each paper introduces a new or modified layer concept. Second, “reasoning layer” may be misread as a description of LLM inference activity, when in the trilogy it names governed substrate content—what the LLM reasons *with*, not the LLM’s reasoning itself. This note establishes the following: all four layer terms carry consistent definitions from their introduction through Paper 3; Papers 1 and 2 establish the definitions; Paper 3 applies them unchanged and adds one property—that only the reasoning layer (DNA layer + action layer) may cross the inter-Self perimeter during Full Aspect Integration, while the instinct layer cannot. Exchange bounding is the governance mechanism enforcing this property.

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1. Why disambiguation is needed

The trilogy accumulates layer vocabulary progressively. Paper 1 does not use the terms “instinct layer,” “DNA layer,” “action layer,” or “reasoning layer” explicitly, but establishes the two-type substrate content structure those terms will name: authored governance specifications that govern the LLM's substrate interactions, and operational records of what the substrate has done. Paper 2 names all four terms explicitly and defines them. Paper 3 uses all four terms with exactly the Paper 2 definitions, applying them at inter-Self scope.

The progressive introduction creates a surface-level ambiguity risk. A reader who encounters "reasoning layer" in Paper 3 without having tracked its Paper 2 definition may read it as a description of how the LLM reasons—as an inference layer. That reading is incorrect and consequential: it mislocates the boundary between what the LLM does and what the substrate does, which is the architectural spine the trilogy defends. A second reader may treat the four layer terms as Paper-2-only vocabulary with no Paper 1 roots, missing the implicit two-type structure Paper 1 establishes and therefore treating the layer architecture as a Paper 2 novelty rather than as a naming of Paper 1 commitments. This note closes both gaps.

2. Paper 1: the implicit two-type content structure

Paper 1 does not use the four layer terms but establishes their architectural basis. The cell—Paper 1's atomic coordination unit—contains two distinct types of substrate content:

> **Authored governance specifications.** Orchestration rules, content-domain specifications, and coordination rules that govern how the cell behaves when it executes over substrate content. These are written by humans (or drafted by LLMs under human authority) and are what governance authors and can modify through directed selection. This is the content Paper 2 will name the *DNA layer*.
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> **Operational records.** The substrate record of what the cell actually did—task instances, outputs, decisions, conflicts surfaced. This is the evidence base that closing a feedback loop requires. This is the content Paper 2 will name the *action layer*.

Paper 1 also establishes the two-component structure of the cell itself: the LLM and the governance substrate, divided at the governance boundary rather than the capability boundary. The LLM handles fast-pattern, high-dimensional inference. The substrate handles coordination, governance, and structured record. This division is the architectural basis for what Paper 2 will name the *instinct layer* and the *reasoning layer*.

Paper 1 does not name these layers because it operates at cell scope, where the division is sufficiently described by the substrate/LLM framing. Paper 2 names them because multi-level composition requires precise vocabulary for what evolves independently and what exchanges across coordination events.

3. Paper 2: four layer terms named

Paper 2 introduces all four terms as explicit vocabulary in the context of the three-level architecture (cell, aspect, Self) and the three evolution mechanisms (instinct evolution, DNA evolution, action-feedback evolution).

Instinct layer. The LLM together with the harness substrate—the organizational character and capability that shapes how the entity interprets and responds to its governance substrate. The instinct layer is not directly authored as governance content. It evolves through LLM upgrades and infrastructure changes, not through directed selection over substrate content. It carries fast-pattern responses that do not require explicit deliberation.

DNA layer. The authored governance specifications at each tier (cell DNA, aspect DNA, Self DNA): orchestration rules, content-domain specifications, coordination rules, harness substrates, verification substrates. This is what governance authors and can modify through directed selection. It is the substrate-side content that specifies how cells, aspects, and Selves behave.

Action layer. The operational records at each tier: what cells and aspects actually did, recorded as substrate content. Task instances, outputs, decisions, and conflict records live here. The action layer is the evidence base for action-feedback evolution—the mechanism by which lived operational experience can feed back into DNA modification under human governance.

Reasoning layer. The collective term for the DNA layer and action layer together. It names the totality of governed substrate content, as distinguished from the instinct layer, which is not governed through substrate content. The reasoning layer is what the LLM reasons *with*—the external, persistent, human-governed substrate that carries coordination knowledge, governance specifications, and operational history. It is not the LLM's reasoning activity.

This last point warrants emphasis. "Reasoning layer" in the trilogy is a substrate-side concept, not a model-side concept. When the trilogy says the reasoning layer is human-governed, it means the substrate content the LLM reads and writes is under human authority—not that human authority governs LLM inference operations directly. The substrate/LLM division that Paper 1 draws at the governance boundary is what makes "reasoning layer" a substrate term: the layer lives outside the model, not inside it.

4. Two categories: the organizing principle

The four layer terms map to two fundamental categories, and naming the categories is what makes the layer vocabulary architecturally coherent rather than merely taxonomic.

Category 1—Instinct layer. Not governed through substrate content. Evolves through LLM upgrade and infrastructure change (instinct evolution), not through directed selection. Does not carry provenance in the substrate-content sense. Cannot, in Paper 3's extension, cross the inter-Self perimeter. The instinct layer is an LLM-side concept that the governance architecture constrains but does not directly author.

Category 2—Reasoning layer (DNA layer + action layer). Governed through substrate content. DNA layer evolves through directed selection under human authority. Action layer feeds action-feedback evolution through the proposal-and-acceptance governance machinery. Both carry provenance. Both can, in Paper 3's extension, cross the inter-Self perimeter when governance authorizes the crossing. The reasoning layer is a substrate-side concept that human governance directly authors and controls.

The categorical distinction is the architectural principle. The four specific layer names supply operational precision within each category: the DNA/action distinction within the reasoning layer specifies what evolves through directed selection versus what evolves through feedback, and what routes to which evolution mechanism when FAI-derived content arrives at a home perimeter. The instinct/reasoning distinction specifies where human governance authority is direct (reasoning layer) and where it is indirect—operating through verification substrates, routing rules, and the orchestration rules that pin high-stakes decisions to the reasoning layer regardless of instinct capability.

5. Paper 3: the cross-perimeter governance addition

Paper 3 applies all four layer definitions unchanged. Its contribution to the layer concepts is a single new property: the specification of which layers may cross the inter-Self perimeter during a Full Aspect Integration (FAI) event.

FAI is Paper 3's canonical operation over the shared substrate—the mechanism through which two or more CKS-governed Selves contribute aspects to a temporarily constructed shared coordination substrate and operate jointly over it. When a Self contributes an aspect to the shared substrate, the contribution surfaces the constituent cells' content, the aspect's own governance specifications, and the aspect's operational records. The question the layer concepts must answer at this scope is: what content is exchangeable, and what is not?

The answer Paper 3 commits to is this: **only DNA-layer and action-layer content—that is, the reasoning layer—may cross the inter-Self perimeter**. LLM weights and instinct-layer content do not exchange across FAI. The instinct layer remains within each Self's home governance perimeter regardless of FAI scope or configuration.

Exchange bounding is the governance mechanism that enforces this property. The exchange-bounding commitment is articulated at FAI mechanism level in Paper 3—it is fresh at inter-Self scope in the sense that prior papers did not need to specify it, because the question of what crosses a perimeter between distinct Selves did not arise at cell scope (Paper 1) or intra-Self scope (Paper 2). But the underlying instinct/reasoning separation that exchange bounding enforces is pure Paper 2 inheritance. Paper 3 does not change the layer definitions; it derives a cross-perimeter property from the definitions that were already in place.

The layer-routing rule at the FAI hand-off boundary follows directly. When a Self ingests FAI-derived content at its home perimeter, the layer of the incoming content determines which Paper 2 evolution mechanism receives it: DNA-layer content routes to DNA evolution operating on home DNA; action-layer content routes to action-feedback evolution operating within home action layers. Instinct evolution receives no FAI input by architectural commitment—there is no pathway from the shared substrate to LLM weights, because instinct-layer content does not cross the inter-Self perimeter in either direction.

6. Prior-art closure

The disambiguation record established by this note forecloses the following adversarial readings:

Foreclosure 1: “Paper 3 introduces new layer concepts.” Paper 3 does not introduce new layer concepts. It applies the four Paper 2 layer definitions at inter-Self scope and adds the cross-perimeter property. The layer definitions themselves are Paper 2 contributions with Paper 1 roots. Any architecture implementing layer-differentiated content at inter-Self scope is operating within territory the trilogy establishes.

Foreclosure 2: “The reasoning layer is a description of LLM inference.” “Reasoning layer” in the trilogy names governed substrate content—the DNA layer and action layer together. It is not a model-side concept and does not describe LLM inference activity. Architectures that use “reasoning layer” to describe LLM inference, inference chains, or chain-of-thought processing are using the term in a different sense. The trilogy’s use is specific and substrate-side.

Foreclosure 3: “Exchange-bounding the instinct layer at inter-Self scope is a novel contribution independent of the layer definitions.” Exchange bounding is a direct consequence of the layer definitions Paper 2 establishes. The instinct layer is not governed through substrate content and does not carry the provenance structure that substrate content carries; it therefore cannot be a unit of authorized exchange across a governed substrate perimeter. The exchange-bounding commitment is the application of the layer architecture to inter-Self scope, not an independent invention.

Foreclosure 4: “The DNA/action distinction within the reasoning layer is arbitrary.” The DNA/action distinction is load-bearing for the three evolution mechanisms. DNA-layer content is what directed selection operates over; action-layer content is what action-feedback evolution operates over. The distinction determines not only what evolves through which mechanism but also what routes to which mechanism when FAI-derived content reaches a home perimeter. Collapsing the two into undifferentiated “substrate content” loses the routing rule and the per-mechanism governance shapes that Paper 2 establishes.

Operational summary

The four layer terms carry these definitions consistently across all three papers:

Term	Category	Paper 1	Paper 2	Paper 3 addition
Instinct layer	Not substrate-governed	Implicit (LLM component)	Named	Cannot cross inter-Self perimeter
DNA layer	Substrate-governed	Implicit (governance specs)	Named	Can cross; routes to DNA evolution at home

Action layer	Substrate-governed	Implicit (operational records)	Named	Can cross; routes to action-feedback at home
Reasoning layer	Substrate-governed (collective)	Implicit (two-type structure)	Named (DNA + action collective)	Exchange currency in FAI

The categorical rule is: reasoning layer (DNA + action) can cross the inter-Self perimeter; instinct layer cannot. Exchange bounding enforces this rule. The layer definitions do not change across papers. Paper 3 derives a cross-perimeter property from definitions already in place.

How to cite this note

Li, W. (2026). Disambiguating "Layer" Across the Trilogy: Four Layer Concepts (Instinct, DNA, Action, Reasoning) Consistently Defined Across Papers—Paper 3 Adds Cross-Perimeter Governance (Reasoning Layer Can Cross; Instinct Layer Cannot). May 15, 2026. ORCID: 0009-0004-8065-3235.